

Fitting and Interpreting Exponential Models

Fitting exponential models to data, reading growth and decay rates, solving for time with logarithms, and rewriting a model with a different base

Directions.

- Every model below states its variable and its units. Answer in the unit the question names.
- Round only at the last step, and only to the precision the question asks for.
- Write the equation you solved, not just the number you got.

1. A savings account is modelled by $A(t) = 1200(1.045)^t$, where A is the balance in dollars and t is the time in years.

(a) The balance at $t = 0$ years, in dollars.

Answer: _____

(b) The annual percent rate of increase.

Answer: _____

2. A lab culture holds 200 cells when the count begins and 3200 cells 4 hours later. The technician models the culture as $N(t) = a b^t$ cells, with t in hours.

(a) Find the hourly growth factor b .

Answer: _____

(b) Use the model to predict the cell count at $t = 7$ hours.

Answer: _____

3. A dose of medication in the bloodstream is modelled by $M(t) = 250(0.86)^t$, where M is in milligrams and t is the time in hours after the dose.

- (a) What percent of the drug present is lost each hour?

Answer: _____

- (b) How much remains after 6 hours, to the nearest 0.1 mg?

Answer: _____ mg

4. A town's population is modelled by $P(t) = 18000(1.023)^t$ people, where t is the number of years after the last census. Find, to the nearest hundredth of a year, how long it takes the population to reach 25000 people.

Answer: _____ years

5. A car is worth 24000 dollars when new and 15360 dollars two years later. Its value is modelled by $V(t) = ab^t$ dollars, with t in years.

(a) Find the yearly decay factor b .

Answer: _____

(b) Predict the value after 5 years, to the nearest dollar.

Answer: _____

6. An investment is modelled by $A(t) = 5000(1.08)^t$ dollars, with t in years. The same investment can be written in continuous form as $A(t) = 5000e^{kt}$.

(a) Find k , to four decimal places.

Answer: _____

(b) Use the continuous form to find the balance after 12 years, to the nearest dollar.

Answer: _____

7. A radioactive tracer decays according to $N(t) = 40(0.5)^{t/6}$, where N is in milligrams and t is the time in hours. How long until 7 mg of the tracer remains? Give the time to the nearest 0.1 hour.

Answer: _____ hours

8. A city's annual water use is modelled by $W(t) = c(0.91)^t$ million litres, where c is the use in the survey year and t is the number of years after it.

(a) By what percent does annual water use fall each year?

Answer: _____

(b) A councillor says use will have fallen by 90 percent after 10 years. Find the actual percent decrease over 10 years, to the nearest 0.1 percent.

Answer: _____

(c) Explain in one or two sentences why the councillor's reasoning fails.

Answer: _____

9. Two analysts model the same fund. Analyst A writes $F(t) = 300(1.06)^{2t}$ dollars; Analyst B writes $F(t) = 300(1.1236)^t$ dollars. In both, t is the time in years.

- (a) Rewrite Analyst A's model with a single yearly growth factor.

Answer: _____

- (b) State Analyst B's annual percent growth rate.

Answer: _____

- (c) Justify in one or two sentences why the two models agree for every t .

Answer: _____

10. A cup of coffee cools according to $T(t) = 21 + 62(0.94)^t$, where T is the temperature in degrees Celsius and t is the time in minutes after pouring. The room stays at 21 degrees Celsius.

- (a) Find the temperature at the moment the coffee is poured.

Answer: _____ degrees

- (b) Find how long the coffee takes to cool to 45 degrees Celsius, to the nearest 0.1 minute.

Answer: _____ minutes